

Data-Driven Classification of U.S. Unemployment Regimes: An Application of Feature Selection and Ensemble Learning on FRED-MD

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Abstract

This study applies machine learning techniques to classify U.S. unemployment regimes using macroeconomic indicators from the Federal Reserve Bank of St. Louis's FRED-MD database. Five feature-selection algorithms—Information Gain, Gain Ratio, ReliefF, Symmetrical Uncertainty, and a manually curated subset—were paired with four classifiers (Random Forest, J48, Logistic Regression, and OneR) to evaluate predictive performance across twenty model combinations. After extensive testing, the **ReliefF + Random Forest** model achieved the highest performance, with **94.38% accuracy** and **Kappa = 0.925**, accurately distinguishing between four unemployment-level categories. The model effectively captured nonlinear relationships and interdependencies among labor, inflation, and monetary variables, aligning with economic theory while maintaining interpretability. These findings suggest that ensemble-based, data-driven approaches can complement traditional econometric tools by providing timely, high-precision insights into labor-market dynamics and cyclical economic transitions.

Introduction

In 2016, McCracken and Ng at the Federal Reserve Bank of St. Louis introduced the **FRED-MD** database—a standardized, monthly macroeconomic dataset that consolidated decades of economic indicators for empirical modeling and forecasting. Their work transformed how researchers approached macroeconomic prediction by providing a consistent, curated set of variables that captured the full breadth of U.S. economic activity, from employment and output to inflation and interest rates. Since then, the FRED-MD dataset has become a benchmark for machine learning and econometric studies exploring business cycles, recession forecasting, and policy analysis.

Building on that foundation, this study applies modern classification methods to predict **unemployment rate categories** from FRED-MD-style indicators. The motivation stems from both the recurring shocks that have reshaped the U.S. economy—the 1980s stagflation, the 2008 financial crisis, and the 2020 COVID-19 recession—and the growing interest in how machine learning can detect early signs of labor-market stress. By combining attribute selection algorithms with multiple classifiers, this project explores how data-driven models can classify economic states with high precision while retaining interpretability. In doing so, it extends the

spirit of McCracken and Ng’s work into a modern, algorithmic framework for understanding the structure and transitions of the macroeconomy.

Methodology

Dataset Overview

This study used a macroeconomic dataset titled *2025-08-MD.csv*, consisting of 800 monthly observations from January 1959 to August 2025. Each instance represented a snapshot of U.S. economic conditions compiled from the Federal Reserve Economic Data (FRED). The dataset initially contained 127 attributes reflecting major economic indicators such as employment, industrial production, money supply, interest rates, and consumer spending. The target variable for prediction was the unemployment rate (**UNRATE**), which was converted into categorical classes for classification analysis. The date variable (**sasdate**) was excluded from the feature set because it carried no predictive value.

Exploratory Analysis

Before preprocessing, basic exploratory analysis was conducted using WEKA’s visualization tools and supplementary time-series plots. The unemployment rate (**UNRATE**) displayed clear cyclical fluctuations, with major peaks aligning with historical recessions such as the early 1980s downturn, the 2008 financial crisis, and the 2020 COVID-19 recession. Preliminary correlation checks revealed that variables tied to employment, consumer prices, and monetary aggregates exhibited strong relationships with unemployment levels. Indicators such as **CES employment series**, **CPIAUCSL** (Consumer Price Index), and **M2SL** (Money Supply) appeared particularly predictive, suggesting that labor and price trends are tightly linked to cyclical unemployment behavior.

Data Preprocessing

Missing values in the dataset were handled using WEKA’s *ReplaceMissingValues* filter, ensuring that all attributes contained complete information. No transformations, scaling, or standardization were applied — all variables were used in their original numeric form. This choice preserved the interpretability of the raw economic indicators and reflected their natural magnitudes.

The target variable **UNRATE** was discretized into four equal-frequency bins using WEKA’s *Discretize* filter. This process converted the continuous unemployment rate into categorical

levels, enabling classification modeling. The bins were defined as follows: $(-\infty, 4.65]$ (*Low*), $(4.65, 5.65]$ (*Moderate-Low*), $(5.65, 6.95]$ (*Moderate-High*), and $(6.95, \infty)$ (*High*).

The dataset was divided into training and testing sets using **stratified random sampling**, ensuring that each class label was proportionally represented in both subsets. An **80/20 split** was applied, with 80% of instances allocated to training and 20% reserved for testing. Stratification maintained class balance, preventing overrepresentation of more common unemployment levels and yielding a more reliable assessment of model generalization.

Attribute Selection

To identify the most relevant predictors of unemployment, four attribute selection algorithms were applied using WEKA's *Select Attributes* interface:

1. **Information Gain (InfoGainAttributeEval)**
2. **Gain Ratio (GainRatioAttributeEval)**
3. **Symmetrical Uncertainty (SymmetricalUncertAttributeEval)**
4. **ReliefF (ReliefFAttributeEval)**

Each algorithm evaluated the relationship between individual attributes and the target variable (**UNRATE**).

- **Information Gain, Gain Ratio, and Symmetrical Uncertainty** produced highly similar rankings, consistently identifying attributes such as *CES3000000008*, *CUSR0000SAS*, *M2SL*, *CPIMEDSL*, *PCEPI*, and *CPIAUCSL* as the most informative. These indicators capture variations in employment, consumer prices, and production — all fundamental components of unemployment cycles.
- **ReliefF**, which uses nearest-neighbor comparisons to assess local dependencies, highlighted a different subset emphasizing short-term labor and financial market indicators, including *UEMPMPAN*, *HWIURATIO*, *BAAFFM*, and *T10YFFM*.

This multi-method approach ensured both linear and non-linear associations with unemployment were captured.

Manual Attribute Subset

After examining all four attribute selection outputs, a **manual feature subset** was constructed to integrate statistical relevance with economic interpretability. Attributes were selected if they (a) appeared consistently across multiple selection algorithms, or (b) represented a distinct macroeconomic domain historically tied to labor dynamics.

Information Gain (25)	Gain Ratio (25)	Symmetrical Uncertainty (25)	ReliefF (25)	Manual Set (25)
CES3000000008	CES3000000008	CES3000000008	UEMPMPA N	RPI,
CUSR0000SAS	CUSR0000SAS	CUSR0000SAS	HWIURATI O	DPCERA3M086S
M2SL	M2SL	M2SL	UEMP150V	BEA
DSERRG3M086S BEA	CPIMEDSL	CES0600000008	UEMP270V	RETAILx
CES0600000008	CES0600000008	CPIMEDSL	BAAFFM	CLF16OV
CPIMEDSL	DSERRG3M086 SBEA	DSERRG3M086S BEA	AWOTMAN	SRVPRD
CES2000000008	CES2000000008	CES2000000008	AAAFFM	AMDMUOx
PCEPI	PCEPI	PCEPI	PERMITS	ISRATIOx
CPIAUCSL	CPIAUCSL	CPIAUCSL	T10YFFM	M1SL
CUSR0000SA0L 5	CPIULFSL	CUSR0000SA0L 5	S&P Dividend Yield	M2REAL
CPIULFSL	CUSR0000SA0L 5	CPIULFSL	PERMIT	BUSLOANS
REALLN	REALLN	REALLN	HOUSTW	WPSFD49502
BOXMBASE	M1SL	M1SL	T5YFFM	OILPRICEx
CUSR0000SA0L 2	CLF16OV	CLF16OV	TB6MS	CPIAPPSL
	CLAIMSx	INVEST		CPITRNSL

M1SL	NONREVSL	CUSR0000SA0L2	PERMITW	CPIMEDSL
CLF16OV	BOXMBASE	RETAILx	EXSZUSx	CUSR0000SAC
INVEST	INVEST	DPCERA3M086S	HOUSTS	CUSR0000SAD
RETAILx	CUSR0000SA0L2	BEA	TB3MS	CUSR0000SAS
DPCERA3M086S	DPCECRA3M086SBEA	BUSINVx	GS1	CUSR0000SA0L2
BEA		RPI	IPDCONGD	CUSR0000SA0L5
RPI				PCEPI
				DDURRG3M086
				SBEA
				CES0600000008
				INVEST
				UNRATE

Modeling and Evaluation

Four supervised classification algorithms were used to model the relationship between macroeconomic indicators and unemployment categories:

1. **Random Forest (weka.classifiers.trees.RandomForest)** – An ensemble of decision trees trained on random subsets of data to enhance robustness and reduce overfitting.
2. **Logistic Regression (weka.classifiers.functions.Logistic)** – A probabilistic model estimating categorical outcomes using a logistic function.
3. **OneR (weka.classifiers.rules.OneR)** – A simple rule-based classifier that provides an interpretable baseline based on a single best attribute.

4. **J48 (weka.classifiers.trees.J48)** – WEKA’s implementation of the C4.5 decision tree algorithm, capable of handling both numeric and categorical attributes.

Each classifier was trained using the 80% stratified training set and evaluated on the 20% test set. Additionally, 10-fold cross-validation was applied to the training data to internally validate model performance. Evaluation metrics included classification accuracy, precision, recall, F-measure, and confusion matrices, allowing direct comparison of classifier effectiveness across different attribute selection methods and the manual subset.

We selected **Random Forest, Logistic Regression, OneR, and J48** because they collectively balance interpretability, flexibility, and robustness across both linear and nonlinear relationships. In contrast to models like **Naive Bayes** and other algorithms that assume independence among features, these classifiers can effectively capture the interdependencies between economic indicators such as employment, inflation, and production, which are inherently correlated in real-world macroeconomic data.

Distribution Shift and Data Splitting

The dataset was first split chronologically to simulate real forecasting, but this caused extreme **distribution shift**: the training data mostly represented low-unemployment decades, while the test set captured volatile periods like 2008 and 2020. Models overfit to one regime and collapsed to single-class predictions with accuracies near 0–3%. To fix this, we switched to **stratified 80/20 sampling**, ensuring all unemployment categories were represented in both training and testing data. This approach produced far more balanced and interpretable results.

Discretization and Binning

The original **equal-width, three-bin** discretization produced heavily skewed classes, with most data falling into a single bin. Switching to **equal-frequency binning** balanced the categories and improved learning stability. Expanding from **three to four bins** added finer distinctions between moderate and elevated unemployment, aligning better with real-world economic conditions. The improvement is clearly shown in the **bar charts** comparing the original skewed distribution to the more even four-bin version.

Manual vs. Automatic Cut Points

Manual bin thresholds were briefly considered for interpretability (e.g., labeling “recession-level” unemployment) but were too rigid for decades of changing baselines.

Automated **equal-frequency discretization** adapted better to modern conditions and provided reproducible, balanced class divisions across all time periods.

Results and Model Performance Analysis

Each attribute-selection method was paired with four classifiers (Random Forest, J48, Logistic Regression, OneR). All models were evaluated on the same 160-instance stratified test set, using identical categorical unemployment classes. Below, each subsection summarizes the main performance metrics—accuracy, precision, recall (sensitivity), specificity, F-measure—and interprets the confusion-matrix behavior.

4.1 Information Gain

Model	Accuracy	Kappa	Mean Abs Err	ROC Area	PRC Area
Random Forest	91.25 %	0.883	0.098	0.989	0.972
J48	91.88 %	0.892	0.097	0.987	0.973
Logistic Regression	88.13 %	0.841	0.129	0.956	0.931
OneR	88.75 %	0.850	0.123	0.950	0.912

Interpretation

- **Sensitivity (Recall):** Both J48 and RF correctly classified ~92–94 % of “Low” and “High” unemployment bins; mid bins slightly lower (~88 – 90 %), showing only minor confusion at regime boundaries.

- **Specificity:** Very high (> 0.95) for all bins; misclassifications were balanced rather than systematic.
- **Precision & F1:** Each around 0.90 – 0.92, confirming consistent prediction quality across all classes.
- **Confusion pattern:** Occasional swaps between neighboring middle classes (4.65–6.95 %), none between extremes.
Conclusion: Information Gain produced stable, interpretable performance with balanced sensitivity/specificity; J48 marginally led overall.

4.2 Gain Ratio

Model	Accuracy	Kappa	MAE	ROC Area	PRC Area
Random Forest	92.50 %	0.900	0.088	0.992	0.978
Logistic Regression	91.88 %	0.892	0.097	0.974	0.952
J48	88.75 %	0.850	0.122	0.951	0.927
OneR	88.75 %	0.850	0.123	0.949	0.914

Interpretation

- **Precision:** RF ≈ 0.93 across all bins; Logistic ≈ 0.91 .
- **Recall (Sensitivity):** RF captured $> 90\%$ of middle-bin cases—the most difficult class.
- **Specificity:** All models ≥ 0.95 ; almost no false alarms in extreme bins.

- **F-measure:** RF = 0.92 average, confirming both low bias and variance.
 - **Confusion Matrix:** Small symmetric off-diagonal counts → no class bias.
Conclusion: Gain Ratio + RF offered an optimal precision-recall trade-off and superior generalization, outperforming both simpler trees and linear regression.
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4.3 ReliefF

Model	Accuracy	Kappa	MAE	ROC Area	PRC Area
Random Forest	94.38 %	0.925	0.066	0.994	0.983
J48	91.25 %	0.883	0.097	0.981	0.961
Logistic Regression	91.25 %	0.883	0.099	0.965	0.949
OneR	65.00 %	0.463	0.258	0.721	0.624

Interpretation

- **Precision:** RF reached ≈ 0.95 overall; Logistic/J48 ≈ 0.91 .
- **Recall/Sensitivity:** RF ≈ 0.94 (average), strongest recall among all methods.
- **Specificity:** Also > 0.96 , confirming extremely low false-positive rates.
- **F1-score:** RF ≈ 0.94 aggregate, the best harmonic balance of precision & recall.
- **Confusion Matrix:** Perfect classification of extreme bins; minor off-by-one errors in middle bins only.
Conclusion: ReliefF's local-sensitivity ranking paired with Random Forest's ensemble

learning yielded the most balanced and accurate model overall.

4.4 Symmetrical Uncertainty

Model	Accuracy	Kappa	MAE	ROC Area	PRC Area
Random Forest	93.75 %	0.918	0.077	0.991	0.979
J48	91.88 %	0.892	0.097	0.981	0.962
Logistic Regression	87.50 %	0.829	0.135	0.949	0.926
OneR	89.38 %	0.852	0.124	0.956	0.930

Interpretation

- **Precision / Recall:** RF $\approx$ 0.93–0.94 for all bins, F1 $\approx$ 0.93; J48 slightly lower ($\approx$ 0.91).
- **Specificity:** Nearly perfect for extreme bins, small mid-class overlap.
- **Confusion Matrix:** No bias—errors evenly distributed; mid bins had 1-to-2 mislabels at most.
Conclusion: Symmetrical Uncertainty handled redundant but correlated features well, giving consistent high performance without overfitting.

4.5 Manual Subset

Model	Accuracy	Kappa	MAE	ROC Area	PRC Area
Random Forest	92.50 %	0.900	0.088	0.990	0.977
J48	90.63 %	0.875	0.103	0.974	0.951
Logistic Regression	89.38 %	0.857	0.113	0.959	0.937
OneR	88.75 %	0.850	0.123	0.952	0.923

Interpretation

- **Precision/Recall:** RF $\approx$ 0.92; J48 $\approx$ 0.90 balanced.
 - **Specificity:** $>$ 0.94 overall.
 - **F1:** RF $\approx$ 0.91; J48 $\approx$ 0.89.
 - **Confusion Matrix:** Slight blending between the two middle bins; otherwise stable.
Conclusion: The manual attribute mix captured economically meaningful signals (labor, prices, liquidity), performing on par with Gain Ratio while remaining interpretable.
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4.6 Cross-Method Comparison

Attribute Selector	Best Model	Accuracy	Kappa	Notes
Information Gain	J48	91.88 %	0.892	Stable, balanced; slight mid-bin confusion
Gain Ratio	Random Forest	92.50 %	0.900	Clean separations; high precision
ReliefF	Random Forest	94.38 %	0.925	Best recall, minimal errors
Symmetrical Uncertainty	Random Forest	93.75 %	0.918	High stability, low redundancy bias
Manual Subset	Random Forest	92.50 %	0.900	Interpretability + accuracy

Trend summary

- **Random Forest** dominates four of five sets, showing the strongest precision–recall–specificity balance.
 - **J48** follows closely, trading a few points of accuracy for simpler interpretability.
 - **Logistic Regression** remains robust but limited in handling nonlinear thresholds.
 - **OneR** confirms attribute relevance yet misses cross-feature interactions.
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4.7 Gold-Star Model: ReliefF + Random Forest

Metrics: Accuracy = 94.38 %, Kappa = 0.925, Precision $\approx$ 0.95, Recall $\approx$ 0.94, Specificity $\approx$ 0.96, F1 $\approx$ 0.94, ROC $\approx$ 0.99.

Interpretation: This model achieved the best overall discrimination with balanced precision and recall across all four unemployment categories, minimal false positives, and high confidence calibration.

Why it excels:

- **ReliefF** highlights features that best separate *neighboring* economic conditions, ideal for macro data where transitions (e.g., mild $\rightarrow$ moderate unemployment) are subtle.
- **Random Forest** aggregates nonlinear relationships among these locally strong features, resisting overfitting and capturing interactions among labor, inflation, and liquidity indicators.
- **Conceptual coherence:** Unemployment regimes in reality emerge from combined movements in production, money supply, and consumer prices. The model's joint, context-aware learning mirrors this dynamic, explaining its strong empirical and theoretical fit.

In summary, while several models reached accuracies above 90 %, **ReliefF + Random Forest** delivered the highest and most balanced predictive power—confirming that feature locality and ensemble learning together best capture the multivariate structure of the U.S. macroeconomy.

Discussion

Across the twenty model runs, several consistent patterns emerged regarding how attribute selection and classifier choice shaped performance. Tree-based models—**Random Forest** and **J48**—outperformed the others because they captured **nonlinear, joint movements** across employment, price, and monetary indicators that define shifts in unemployment regimes. **Logistic Regression** remained strong for broad patterns but lacked flexibility to handle interactions or thresholds, while **OneR** performed acceptably only when a single dominant feature explained much of the variance.

At a finer level, the **per-class metrics** reveal how different algorithms handled distinct segments of the unemployment distribution. The extreme bins (very low and very high unemployment) were the easiest to classify, achieving near-perfect **recall and specificity** since economic

indicators tend to move synchronously during recessions or booms. The most challenging range was the middle of the distribution—months with moderate or borderline unemployment—where weaker models confused adjacent categories. The best-performing models, particularly **ReliefF + Random Forest**, preserved clear boundaries between these middle classes, posting balanced **precision**, **recall**, and **F1 scores** above 0.9 and minimizing off-by-one errors in their confusion matrices. Errors that did occur were typically **ordinally local**, meaning the model confused neighboring bins rather than extremes—an expected and economically reasonable pattern.

Feature-selection algorithms also revealed distinct strengths. **Information Gain**, **Gain Ratio**, and **Symmetrical Uncertainty** prioritized consistent macroeconomic signals such as consumer prices, employment, and money supply, producing high and stable accuracy across all models. **ReliefF**, in contrast, rewarded locally discriminative features—variables that differentiate nearby cases—allowing models to recognize subtle transitions between moderate and elevated unemployment. This property explains its superior performance when paired with Random Forest, which can exploit localized relationships through ensemble averaging. The **manual attribute subset** performed nearly as well as automated methods, validating that domain expertise can rival algorithmic feature ranking when grounded in sound economic logic.

Model-wise, **Random Forest** consistently provided the best balance of sensitivity and specificity. Its ensemble structure captured complex, nonlinear dependencies and avoided overfitting by averaging across many decorrelated decision trees. **J48** traded a small loss in accuracy for interpretability, with tree branches corresponding to understandable thresholds in inflation or employment variables. **Logistic Regression** achieved good average precision but underfit subtle regime boundaries, while **OneR** confirmed that many variables were individually predictive even if it failed to model interactions.

Ultimately, the **ReliefF + Random Forest** model stands out as the gold standard. It reached **94.38% accuracy** and a **Kappa of 0.925**, combining the highest predictive balance with robust conceptual grounding. ReliefF identified attributes that best separate nearby economic conditions, and Random Forest aggregated them into stable nonlinear decision surfaces. Together, they produced high **sensitivity** (capturing nearly all true cases), strong **specificity** (avoiding false positives), and the best **F1-scores** across every unemployment bin. Conceptually, this makes sense: unemployment is not driven by a single factor but by **interacting trends** in prices, output, and liquidity. A model that integrates local feature relevance with nonlinear decision boundaries mirrors this real-world complexity.

The results confirm that combining **feature-level locality** with **ensemble modeling** yields not just superior metrics but a framework that aligns statistically and economically with how the labor market behaves.

Limitations

While the models achieved high predictive accuracy, several limitations constrain the generalizability of these results.

First, the dataset represents aggregated **macroeconomic indicators**, which evolve under shifting policy and structural conditions. Relationships that held historically—such as between unemployment and inflation—may weaken or invert in future periods, reducing the model’s long-term reliability.

Second, the **discretization of UNRATE** into categorical bins, while necessary for classification, inherently loses granularity. Small yet economically significant changes in unemployment (e.g., 5.9% vs. 6.1%) are treated as distinct classes, potentially overstating model precision.

Third, **class imbalance and temporal dependence** remain partial concerns. Although stratified sampling improved representation, macroeconomic data are autocorrelated, meaning that nearby observations are not truly independent—something Random Forests and J48 cannot explicitly model.

Finally, the study relies on **static attribute selection and snapshot evaluation**. Real-world forecasting would require continuous retraining and feature re-evaluation as new data arrive. Future work could incorporate time-aware models, cross-validation across rolling windows, and additional socioeconomic variables to enhance generalizability.

Conclusion

This study demonstrates that machine learning models can effectively classify unemployment regimes using macroeconomic indicators, with accuracies consistently above 90%. Among the twenty models tested, the **Relieff + Random Forest** combination emerged as the most accurate and conceptually sound, capturing nonlinear relationships and subtle transitions between economic states. The results highlight that unemployment levels are best understood through interconnected variables—employment, prices, and money supply—rather than isolated indicators. While discretization and temporal dependence limit perfect generalization, the model’s performance suggests strong potential for data-driven analysis of labor-market trends. Overall, this work reinforces that ensemble methods grounded in economically meaningful features can bridge the gap between statistical modeling and real-world macroeconomic behavior.

Future Work

Future research can extend this study in several directions. First, incorporating **time-series-aware models** such as LSTM or Temporal Random Forests could capture the sequential dependencies between months that static classifiers overlook. Second, refining **feature selection dynamically**—updating attribute importance as new data are released—would better reflect evolving economic relationships. Third, experimenting with **hybrid regression-classification frameworks** could preserve continuous unemployment information while still identifying distinct economic regimes. In addition, integrating **external sentiment or policy variables** (e.g., consumer confidence indices or fiscal policy measures) may further enhance predictive accuracy. Broadly, the model could evolve into a **real-time economic monitoring system**, continuously retraining on new FRED data to provide adaptive, interpretable forecasts for policymakers and analysts.

References

McCracken, Michael W., and Ng, Serena (2016). “FRED-MD: A Monthly Database for Macroeconomic Research.” Federal Reserve Bank of St. Louis. Data retrieved via FRED Database.

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Results of All Trial Runs

InfoGain RandomForest	Time taken to build model: 0.1 seconds								
	=== Evaluation on test set ===								
	Time taken to test model on supplied test set: 0.01 seconds								
	=== Summary ===								
	Correctly Classified Instances	146		91.25	%				
	Incorrectly Classified Instances	14		8.75	%				
	Kappa statistic	0.8832							
	Mean absolute error	0.0568							
	Root mean squared error	0.1679							
	Relative absolute error	15.1546	%						
	Root relative squared error	38.7846	%						
	Total Number of Instances	160							
	=== Detailed Accuracy By Class ===								
		TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
	Class								

	<table><tr><td>0.951</td><td>0.017</td><td>0.951</td><td>0.951</td><td>0.951</td><td>0.934</td><td>0.985</td><td>0.980</td><td>'(-inf-4.65]'</td></tr><tr><td>0.884</td><td>0.051</td><td>0.864</td><td>0.884</td><td>0.874</td><td>0.826</td><td>0.985</td><td>0.937</td><td>'(4.65-5.65]'</td></tr><tr><td>0.838</td><td>0.041</td><td>0.861</td><td>0.838</td><td>0.849</td><td>0.805</td><td>0.973</td><td>0.945</td><td>'(5.65-6.95]'</td></tr><tr><td>0.974</td><td>0.008</td><td>0.974</td><td>0.974</td><td>0.974</td><td>0.966</td><td>0.999</td><td>0.997</td><td>'(6.95-inf)'</td></tr><tr><td>Weighted Avg.</td><td>0.913</td><td>0.030</td><td>0.912</td><td>0.913</td><td>0.912</td><td>0.883</td><td>0.985</td><td>0.964</td></tr></table> === Confusion Matrix === a b c d <-- classified as 39 1 1 0 a = '(-inf-4.65]' 2 38 3 0 b = '(4.65-5.65]' 0 5 31 1 c = '(5.65-6.95]' 0 0 1 38 d = '(6.95-inf)'	0.951	0.017	0.951	0.951	0.951	0.934	0.985	0.980	'(-inf-4.65]'	0.884	0.051	0.864	0.884	0.874	0.826	0.985	0.937	'(4.65-5.65]'	0.838	0.041	0.861	0.838	0.849	0.805	0.973	0.945	'(5.65-6.95]'	0.974	0.008	0.974	0.974	0.974	0.966	0.999	0.997	'(6.95-inf)'	Weighted Avg.	0.913	0.030	0.912	0.913	0.912	0.883	0.985	0.964																																	
0.951	0.017	0.951	0.951	0.951	0.934	0.985	0.980	'(-inf-4.65]'																																																																							
0.884	0.051	0.864	0.884	0.874	0.826	0.985	0.937	'(4.65-5.65]'																																																																							
0.838	0.041	0.861	0.838	0.849	0.805	0.973	0.945	'(5.65-6.95]'																																																																							
0.974	0.008	0.974	0.974	0.974	0.966	0.999	0.997	'(6.95-inf)'																																																																							
Weighted Avg.	0.913	0.030	0.912	0.913	0.912	0.883	0.985	0.964																																																																							
InfoGain Logistic	Time taken to build model: 7.95 seconds === Evaluation on test set === Time taken to test model on supplied test set: 0 seconds === Summary === <table><tr><td>Correctly Classified Instances</td><td>141</td><td>88.125 %</td></tr><tr><td>Incorrectly Classified Instances</td><td>19</td><td>11.875 %</td></tr><tr><td>Kappa statistic</td><td>0.8416</td><td></td></tr><tr><td>Mean absolute error</td><td>0.0614</td><td></td></tr><tr><td>Root mean squared error</td><td>0.2092</td><td></td></tr><tr><td>Relative absolute error</td><td>16.3965 %</td><td></td></tr><tr><td>Root relative squared error</td><td>48.3292 %</td><td></td></tr><tr><td>Total Number of Instances</td><td>160</td><td></td></tr></table> === Detailed Accuracy By Class === <table><tr><td>Class</td><td>TP Rate</td><td>FP Rate</td><td>Precision</td><td>Recall</td><td>F-Measure</td><td>MCC</td><td>ROC Area</td><td>PRC Area</td></tr><tr><td></td><td>0.951</td><td>0.034</td><td>0.907</td><td>0.951</td><td>0.929</td><td>0.904</td><td>0.970</td><td>0.968</td></tr><tr><td></td><td>0.767</td><td>0.060</td><td>0.825</td><td>0.767</td><td>0.795</td><td>0.724</td><td>0.964</td><td>0.896</td></tr><tr><td></td><td>0.838</td><td>0.049</td><td>0.838</td><td>0.838</td><td>0.838</td><td>0.789</td><td>0.978</td><td>0.931</td></tr><tr><td></td><td>0.974</td><td>0.017</td><td>0.950</td><td>0.974</td><td>0.962</td><td>0.950</td><td>0.992</td><td>0.951</td></tr><tr><td>Weighted Avg.</td><td>0.881</td><td>0.040</td><td>0.879</td><td>0.881</td><td>0.880</td><td>0.840</td><td>0.975</td><td>0.936</td></tr></table> === Confusion Matrix === a b c d <-- classified as 39 1 0 1 a = '(-inf-4.65]' 4 33 6 0 b = '(4.65-5.65]' 0 5 31 1 c = '(5.65-6.95]' 0 1 0 38 d = '(6.95-inf)'	Correctly Classified Instances	141	88.125 %	Incorrectly Classified Instances	19	11.875 %	Kappa statistic	0.8416		Mean absolute error	0.0614		Root mean squared error	0.2092		Relative absolute error	16.3965 %		Root relative squared error	48.3292 %		Total Number of Instances	160		Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area		0.951	0.034	0.907	0.951	0.929	0.904	0.970	0.968		0.767	0.060	0.825	0.767	0.795	0.724	0.964	0.896		0.838	0.049	0.838	0.838	0.838	0.789	0.978	0.931		0.974	0.017	0.950	0.974	0.962	0.950	0.992	0.951	Weighted Avg.	0.881	0.040	0.879	0.881	0.880	0.840	0.975	0.936
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InfoGain OneR	Time taken to build model: 0.01 seconds === Evaluation on test set === Time taken to test model on supplied test set: 0 seconds === Summary === <table><tr><td>Correctly Classified Instances</td><td>142</td><td>88.75 %</td></tr><tr><td>Incorrectly Classified Instances</td><td>18</td><td>11.25 %</td></tr><tr><td>Kappa statistic</td><td>0.8497</td><td></td></tr><tr><td>Mean absolute error</td><td>0.0563</td><td></td></tr><tr><td>Root mean squared error</td><td>0.2372</td><td></td></tr><tr><td>Relative absolute error</td><td>15.0128 %</td><td></td></tr><tr><td>Root relative squared error</td><td>54.7992 %</td><td></td></tr><tr><td>Total Number of Instances</td><td>160</td><td></td></tr></table>	Correctly Classified Instances	142	88.75 %	Incorrectly Classified Instances	18	11.25 %	Kappa statistic	0.8497		Mean absolute error	0.0563		Root mean squared error	0.2372		Relative absolute error	15.0128 %		Root relative squared error	54.7992 %		Total Number of Instances	160																																																							
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InfoGain J48	<pre> Time taken to build model: 0.05 seconds ==== Evaluation on test set ==== Time taken to test model on supplied test set: 0 seconds ==== Summary ==== Correctly Classified Instances 147 91.875 % Incorrectly Classified Instances 13 8.125 % Kappa statistic 0.8915 Mean absolute error 0.0447 Root mean squared error 0.1969 Relative absolute error 11.9214 % Root relative squared error 45.4999 % Total Number of Instances 160 ==== Detailed Accuracy By Class ==== TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class 0.902 0.000 1.000 0.902 0.949 0.934 0.962 0.946 '(-inf-4.65]' 0.930 0.051 0.870 0.930 0.899 0.861 0.937 0.822 '(4.65-5.65]' 0.892 0.041 0.868 0.892 0.880 0.843 0.916 0.784 '(5.65-6.95]' 0.949 0.017 0.949 0.949 0.949 0.932 0.977 0.924 '(6.95-inf)' Weighted Avg. 0.919 0.027 0.922 0.919 0.919 0.893 0.948 0.870 ==== Confusion Matrix ==== a b c d <-- classified as 37 3 0 1 a = '(-inf-4.65]' 0 40 3 0 b = '(4.65-5.65]' 0 3 33 1 c = '(5.65-6.95]' 0 0 2 37 d = '(6.95-inf)' </pre>
GainRatio RandomForest	<pre> Time taken to build model: 0.09 seconds ==== Evaluation on test set ==== Time taken to test model on supplied test set: 0 seconds ==== Summary ==== Correctly Classified Instances 148 92.5 % Incorrectly Classified Instances 12 7.5 % Kappa statistic 0.8999 Mean absolute error 0.0537 </pre>

	Root mean squared error0.1515 Relative absolute error14.3206 % Root relative squared error34.9991 % Total Number of Instances160 === Detailed Accuracy By Class === <table><thead><tr><th></th><th>TP Rate</th><th>FP Rate</th><th>Precision</th><th>Recall</th><th>F-Measure</th><th>MCC</th><th>ROC Area</th><th>PRC Area</th></tr></thead><tbody><tr><td>Class</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>0.927</td><td>0.008</td><td>0.974</td><td>0.927</td><td>0.950</td><td>0.934</td><td>0.998</td><td>'(-inf-4.65]'</td></tr><tr><td></td><td>0.907</td><td>0.043</td><td>0.886</td><td>0.907</td><td>0.897</td><td>0.858</td><td>0.991</td><td>'(4.65-5.65]'</td></tr><tr><td></td><td>0.892</td><td>0.033</td><td>0.892</td><td>0.892</td><td>0.892</td><td>0.859</td><td>0.991</td><td>'(5.65-6.95]'</td></tr><tr><td></td><td>0.974</td><td>0.017</td><td>0.950</td><td>0.974</td><td>0.962</td><td>0.950</td><td>0.999</td><td>'(6.95-inf)'</td></tr><tr><td>Weighted Avg.</td><td>0.925</td><td>0.025</td><td>0.926</td><td>0.925</td><td>0.925</td><td>0.900</td><td>0.995</td><td>0.985</td></tr></tbody></table> === Confusion Matrix === a b c d <-- classified as 38 3 0 0 a = '(-inf-4.65]' 1 39 3 0 b = '(4.65-5.65]' 0 2 33 2 c = '(5.65-6.95]' 0 0 1 38 d = '(6.95-inf)'		TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class										0.927	0.008	0.974	0.927	0.950	0.934	0.998	'(-inf-4.65]'		0.907	0.043	0.886	0.907	0.897	0.858	0.991	'(4.65-5.65]'		0.892	0.033	0.892	0.892	0.892	0.859	0.991	'(5.65-6.95]'		0.974	0.017	0.950	0.974	0.962	0.950	0.999	'(6.95-inf)'	Weighted Avg.	0.925	0.025	0.926	0.925	0.925	0.900	0.995	0.985																								
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GainRatio Logistic	Time taken to build model: 11.52 seconds === Evaluation on test set === Time taken to test model on supplied test set: 0 seconds === Summary === <table><tbody><tr><td>Correctly Classified Instances</td><td>147</td><td>91.875 %</td></tr><tr><td>Incorrectly Classified Instances</td><td>13</td><td>8.125 %</td></tr><tr><td>Kappa statistic</td><td>0.8915</td><td></td></tr><tr><td>Mean absolute error</td><td>0.0425</td><td></td></tr><tr><td>Root mean squared error</td><td>0.1962</td><td></td></tr><tr><td>Relative absolute error</td><td>11.3499 %</td><td></td></tr><tr><td>Root relative squared error</td><td>45.3316 %</td><td></td></tr><tr><td>Total Number of Instances</td><td>160</td><td></td></tr></tbody></table> === Detailed Accuracy By Class === <table><thead><tr><th></th><th>TP Rate</th><th>FP Rate</th><th>Precision</th><th>Recall</th><th>F-Measure</th><th>MCC</th><th>ROC Area</th><th>PRC Area</th></tr></thead><tbody><tr><td>Class</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>0.951</td><td>0.008</td><td>0.975</td><td>0.951</td><td>0.963</td><td>0.951</td><td>0.993</td><td>'(-inf-4.65]'</td></tr><tr><td></td><td>0.930</td><td>0.051</td><td>0.870</td><td>0.930</td><td>0.899</td><td>0.861</td><td>0.977</td><td>'(4.65-5.65]'</td></tr><tr><td></td><td>0.838</td><td>0.033</td><td>0.886</td><td>0.838</td><td>0.861</td><td>0.821</td><td>0.983</td><td>'(5.65-6.95]'</td></tr><tr><td></td><td>0.949</td><td>0.017</td><td>0.949</td><td>0.949</td><td>0.949</td><td>0.932</td><td>0.997</td><td>'(6.95-inf)'</td></tr><tr><td>Weighted Avg.</td><td>0.919</td><td>0.027</td><td>0.920</td><td>0.919</td><td>0.919</td><td>0.892</td><td>0.987</td><td>0.950</td></tr></tbody></table> === Confusion Matrix === a b c d <-- classified as 39 2 0 0 a = '(-inf-4.65]' 1 40 2 0 b = '(4.65-5.65]' 0 4 31 2 c = '(5.65-6.95]' 0 0 2 37 d = '(6.95-inf)'	Correctly Classified Instances	147	91.875 %	Incorrectly Classified Instances	13	8.125 %	Kappa statistic	0.8915		Mean absolute error	0.0425		Root mean squared error	0.1962		Relative absolute error	11.3499 %		Root relative squared error	45.3316 %		Total Number of Instances	160			TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class										0.951	0.008	0.975	0.951	0.963	0.951	0.993	'(-inf-4.65]'		0.930	0.051	0.870	0.930	0.899	0.861	0.977	'(4.65-5.65]'		0.838	0.033	0.886	0.838	0.861	0.821	0.983	'(5.65-6.95]'		0.949	0.017	0.949	0.949	0.949	0.932	0.997	'(6.95-inf)'	Weighted Avg.	0.919	0.027	0.920	0.919	0.919	0.892	0.987	0.950
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GainRatio J48	<pre> Time taken to build model: 0.02 seconds ==== Evaluation on test set ==== Time taken to test model on supplied test set: 0 seconds ==== Summary ==== Correctly Classified Instances 142 88.75 % Incorrectly Classified Instances 18 11.25 % Kappa statistic 0.8499 Mean absolute error 0.0603 Root mean squared error 0.2272 Relative absolute error 16.0955 % Root relative squared error 52.505 % Total Number of Instances 160 ==== Detailed Accuracy By Class ==== TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class 0.927 0.025 0.927 0.927 0.927 0.902 0.958 0.942 '(-inf-4.65]' 0.814 0.051 0.854 0.814 0.833 0.774 0.928 0.799 '(4.65-5.65]' 0.838 0.057 0.816 0.838 0.827 0.774 0.902 0.761 '(5.65-6.95]' 0.974 0.017 0.950 0.974 0.962 0.950 0.990 0.939 '(6.95-inf)' Weighted Avg. 0.888 0.037 0.887 0.888 0.887 0.850 0.945 0.861 ==== Confusion Matrix ==== a b c d <-- classified as 38 2 1 0 a = '(-inf-4.65]' 3 35 5 0 b = '(4.65-5.65]' 0 4 31 2 c = '(5.65-6.95]' 0 0 1 38 d = '(6.95-inf)' </pre>
ReliefF RandomForest	<pre> Time taken to test model on supplied test set: 0.01 seconds </pre>

	=== Summary === Correctly Classified Instances15194.375 % Incorrectly Classified Instances95.625 % Kappa statistic0.9249 Mean absolute error0.0675 Root mean squared error0.1558 Relative absolute error18.0154 % Root relative squared error35.9906 % Total Number of Instances160 === Detailed Accuracy By Class === <table><thead><tr><th></th><th>TP Rate</th><th>FP Rate</th><th>Precision</th><th>Recall</th><th>F-Measure</th><th>MCC</th><th>ROC Area</th><th>PRC Area</th></tr></thead><tbody><tr><td>Class</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>0.976</td><td>0.008</td><td>0.976</td><td>0.976</td><td>0.976</td><td>0.967</td><td>0.999</td><td>0.997</td></tr><tr><td></td><td>0.953</td><td>0.043</td><td>0.891</td><td>0.953</td><td>0.921</td><td>0.892</td><td>0.989</td><td>0.968</td></tr><tr><td></td><td>0.865</td><td>0.016</td><td>0.941</td><td>0.865</td><td>0.901</td><td>0.875</td><td>0.989</td><td>0.971</td></tr><tr><td></td><td>0.974</td><td>0.008</td><td>0.974</td><td>0.974</td><td>0.974</td><td>0.966</td><td>0.999</td><td>0.997</td></tr><tr><td>Weighted Avg.</td><td>0.944</td><td>0.019</td><td>0.945</td><td>0.944</td><td>0.944</td><td>0.925</td><td>0.994</td><td>0.983</td></tr></tbody></table> === Confusion Matrix === <table><thead><tr><th>a</th><th>b</th><th>c</th><th>d</th><th><-- classified as</th></tr></thead><tbody><tr><td>40</td><td>1</td><td>0</td><td>0</td><td>a = '(-inf-4.65]'</td></tr><tr><td>1</td><td>41</td><td>1</td><td>0</td><td>b = '(4.65-5.65]'</td></tr><tr><td>0</td><td>4</td><td>32</td><td>1</td><td>c = '(5.65-6.95]'</td></tr><tr><td>0</td><td>0</td><td>1</td><td>38</td><td>d = '(6.95-inf)'</td></tr></tbody></table>		TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class										0.976	0.008	0.976	0.976	0.976	0.967	0.999	0.997		0.953	0.043	0.891	0.953	0.921	0.892	0.989	0.968		0.865	0.016	0.941	0.865	0.901	0.875	0.989	0.971		0.974	0.008	0.974	0.974	0.974	0.966	0.999	0.997	Weighted Avg.	0.944	0.019	0.945	0.944	0.944	0.925	0.994	0.983	a	b	c	d	<-- classified as	40	1	0	0	a = '(-inf-4.65]'	1	41	1	0	b = '(4.65-5.65]'	0	4	32	1	c = '(5.65-6.95]'	0	0	1	38	d = '(6.95-inf)'
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ReliefF OneR

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0 seconds

=== Summary ===

Correctly Classified Instances	104	65	%
Incorrectly Classified Instances	56	35	%
Kappa statistic	0.5326		
Mean absolute error	0.175		
Root mean squared error	0.4183		
Relative absolute error	46.7066	%	
Root relative squared error	96.6566	%	
Total Number of Instances	160		

=== Detailed Accuracy By Class ===

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
	0.829	0.084	0.773	0.829	0.800	0.729	0.873	0.685
	0.581	0.128	0.625	0.581	0.602	0.464	0.727	0.476
	0.405	0.106	0.536	0.405	0.462	0.333	0.650	0.355
	0.769	0.149	0.625	0.769	0.690	0.581	0.810	0.537
Weighted Avg.	0.650	0.117	0.642	0.650	0.642	0.530	0.767	0.516

=== Confusion Matrix ===

```

a b c d <-- classified as
34 6 1 0 | a = '(-inf-4.65]'
6 25 5 7 | b = '(4.65-5.65]'
4 7 15 11 | c = '(5.65-6.95]'
0 2 7 30 | d = '(6.95-inf)'

```

ReliefF J48

Time taken to build model: 0.04 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0 seconds

=== Summary ===

Correctly Classified Instances	146	91.25	%
Incorrectly Classified Instances	14	8.75	%
Kappa statistic	0.8832		
Mean absolute error	0.0523		
Root mean squared error	0.2082		
Relative absolute error	13.9526	%	
Root relative squared error	48.1152	%	
Total Number of Instances	160		

=== Detailed Accuracy By Class ===

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
	0.951	0.025	0.929	0.951	0.940	0.919	0.986	0.926
	0.860	0.043	0.881	0.860	0.871	0.824	0.910	0.808
	0.838	0.033	0.886	0.838	0.861	0.821	0.922	0.803
	1.000	0.017	0.951	1.000	0.975	0.967	0.995	0.971
Weighted Avg.	0.913	0.029	0.911	0.913	0.912	0.883	0.953	0.877

=== Confusion Matrix ===

```

a b c d <-- classified as
39 2 0 0 | a = '(-inf-4.65]'
2 37 4 0 | b = '(4.65-5.65]'
1 3 31 2 | c = '(5.65-6.95]'

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	0 0 0 39 d = '(6.95-inf)'																																																																																																																
SymmUncert RandomForest	Time taken to build model: 0.08 seconds === Evaluation on test set === Time taken to test model on supplied test set: 0 seconds === Summary === <table><tr><td>Correctly Classified Instances</td><td>150</td><td>93.75 %</td></tr><tr><td>Incorrectly Classified Instances</td><td>10</td><td>6.25 %</td></tr><tr><td>Kappa statistic</td><td>0.9166</td><td></td></tr><tr><td>Mean absolute error</td><td>0.0546</td><td></td></tr><tr><td>Root mean squared error</td><td>0.1657</td><td></td></tr><tr><td>Relative absolute error</td><td>14.5791 %</td><td></td></tr><tr><td>Root relative squared error</td><td>38.2793 %</td><td></td></tr><tr><td>Total Number of Instances</td><td>160</td><td></td></tr></table> === Detailed Accuracy By Class === <table><tr><td></td><td>TP Rate</td><td>FP Rate</td><td>Precision</td><td>Recall</td><td>F-Measure</td><td>MCC</td><td>ROC Area</td><td>PRC Area</td></tr><tr><td>Class</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>0.951</td><td>0.008</td><td>0.975</td><td>0.951</td><td>0.963</td><td>0.951</td><td>0.996</td><td>0.991</td></tr><tr><td></td><td>0.930</td><td>0.034</td><td>0.909</td><td>0.930</td><td>0.920</td><td>0.890</td><td>0.984</td><td>0.934</td></tr><tr><td></td><td>0.892</td><td>0.033</td><td>0.892</td><td>0.892</td><td>0.892</td><td>0.859</td><td>0.972</td><td>0.945</td></tr><tr><td></td><td>0.974</td><td>0.008</td><td>0.974</td><td>0.974</td><td>0.974</td><td>0.966</td><td>0.999</td><td>0.997</td></tr><tr><td>Weighted Avg.</td><td>0.938</td><td>0.021</td><td>0.938</td><td>0.938</td><td>0.938</td><td>0.917</td><td>0.988</td><td>0.966</td></tr></table> === Confusion Matrix === <table><tr><td>a</td><td>b</td><td>c</td><td>d</td><td><-- classified as</td></tr><tr><td>39</td><td>1</td><td>1</td><td>0</td><td>a = '(-inf-4.65]'</td></tr><tr><td>1</td><td>40</td><td>2</td><td>0</td><td>b = '(4.65-5.65]'</td></tr><tr><td>0</td><td>3</td><td>33</td><td>1</td><td>c = '(5.65-6.95]'</td></tr><tr><td>0</td><td>0</td><td>1</td><td>38</td><td>d = '(6.95-inf)'</td></tr></table>	Correctly Classified Instances	150	93.75 %	Incorrectly Classified Instances	10	6.25 %	Kappa statistic	0.9166		Mean absolute error	0.0546		Root mean squared error	0.1657		Relative absolute error	14.5791 %		Root relative squared error	38.2793 %		Total Number of Instances	160			TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class										0.951	0.008	0.975	0.951	0.963	0.951	0.996	0.991		0.930	0.034	0.909	0.930	0.920	0.890	0.984	0.934		0.892	0.033	0.892	0.892	0.892	0.859	0.972	0.945		0.974	0.008	0.974	0.974	0.974	0.966	0.999	0.997	Weighted Avg.	0.938	0.021	0.938	0.938	0.938	0.917	0.988	0.966	a	b	c	d	<-- classified as	39	1	1	0	a = '(-inf-4.65]'	1	40	2	0	b = '(4.65-5.65]'	0	3	33	1	c = '(5.65-6.95]'	0	0	1	38	d = '(6.95-inf)'
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SymmUncert J48	Time taken to build model: 0.01 seconds === Evaluation on test set === Time taken to test model on supplied test set: 0 seconds === Summary === Correctly Classified Instances14791.875 % Incorrectly Classified Instances138.125 % Kappa statistic0.8915 Mean absolute error0.0447 Root mean squared error0.1969 Relative absolute error11.9214 % Root relative squared error45.4999 % Total Number of Instances160 === Detailed Accuracy By Class ===																																																															

	<table><tr><th></th><th>TP Rate</th><th>FP Rate</th><th>Precision</th><th>Recall</th><th>F-Measure</th><th>MCC</th><th>ROC Area</th><th>PRC Area</th></tr><tr><td>Class</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>0.902</td><td>0.000</td><td>1.000</td><td>0.902</td><td>0.949</td><td>0.934</td><td>0.962</td><td>0.946</td></tr><tr><td></td><td>0.930</td><td>0.051</td><td>0.870</td><td>0.930</td><td>0.899</td><td>0.861</td><td>0.937</td><td>0.822</td></tr><tr><td></td><td>0.892</td><td>0.041</td><td>0.868</td><td>0.892</td><td>0.880</td><td>0.843</td><td>0.916</td><td>0.784</td></tr><tr><td></td><td>0.949</td><td>0.017</td><td>0.949</td><td>0.949</td><td>0.949</td><td>0.932</td><td>0.977</td><td>0.924</td></tr><tr><td>Weighted Avg.</td><td>0.919</td><td>0.027</td><td>0.922</td><td>0.919</td><td>0.919</td><td>0.893</td><td>0.948</td><td>0.870</td></tr></table> === Confusion Matrix === a b c d <-- classified as 37 3 0 1 a = '(-inf-4.65]' 0 40 3 0 b = '(4.65-5.65]' 0 3 33 1 c = '(5.65-6.95]' 0 0 2 37 d = '(6.95-inf)'		TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class										0.902	0.000	1.000	0.902	0.949	0.934	0.962	0.946		0.930	0.051	0.870	0.930	0.899	0.861	0.937	0.822		0.892	0.041	0.868	0.892	0.880	0.843	0.916	0.784		0.949	0.017	0.949	0.949	0.949	0.932	0.977	0.924	Weighted Avg.	0.919	0.027	0.922	0.919	0.919	0.893	0.948	0.870																								
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Manual RandomForest	Time taken to build model: 0.09 seconds === Evaluation on test set === Time taken to test model on supplied test set: 0 seconds === Summary === <table><tr><td>Correctly Classified Instances</td><td>148</td><td>92.5 %</td></tr><tr><td>Incorrectly Classified Instances</td><td>12</td><td>7.5 %</td></tr><tr><td>Kappa statistic</td><td>0.8999</td><td></td></tr><tr><td>Mean absolute error</td><td>0.0595</td><td></td></tr><tr><td>Root mean squared error</td><td>0.1692</td><td></td></tr><tr><td>Relative absolute error</td><td>15.8802 %</td><td></td></tr><tr><td>Root relative squared error</td><td>39.1039 %</td><td></td></tr><tr><td>Total Number of Instances</td><td>160</td><td></td></tr></table> === Detailed Accuracy By Class === <table><tr><th></th><th>TP Rate</th><th>FP Rate</th><th>Precision</th><th>Recall</th><th>F-Measure</th><th>MCC</th><th>ROC Area</th><th>PRC Area</th></tr><tr><td>Class</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td>0.951</td><td>0.008</td><td>0.975</td><td>0.951</td><td>0.963</td><td>0.951</td><td>0.985</td><td>0.980</td></tr><tr><td></td><td>0.930</td><td>0.043</td><td>0.889</td><td>0.930</td><td>0.909</td><td>0.875</td><td>0.985</td><td>0.934</td></tr><tr><td></td><td>0.865</td><td>0.041</td><td>0.865</td><td>0.865</td><td>0.865</td><td>0.824</td><td>0.970</td><td>0.940</td></tr><tr><td></td><td>0.949</td><td>0.008</td><td>0.974</td><td>0.949</td><td>0.961</td><td>0.949</td><td>0.998</td><td>0.995</td></tr><tr><td>Weighted Avg.</td><td>0.925</td><td>0.025</td><td>0.926</td><td>0.925</td><td>0.925</td><td>0.901</td><td>0.985</td><td>0.962</td></tr></table> === Confusion Matrix === a b c d <-- classified as 39 1 1 0 a = '(-inf-4.65]' 1 40 2 0 b = '(4.65-5.65]' 0 4 32 1 c = '(5.65-6.95]' 0 0 2 37 d = '(6.95-inf)'	Correctly Classified Instances	148	92.5 %	Incorrectly Classified Instances	12	7.5 %	Kappa statistic	0.8999		Mean absolute error	0.0595		Root mean squared error	0.1692		Relative absolute error	15.8802 %		Root relative squared error	39.1039 %		Total Number of Instances	160			TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class										0.951	0.008	0.975	0.951	0.963	0.951	0.985	0.980		0.930	0.043	0.889	0.930	0.909	0.875	0.985	0.934		0.865	0.041	0.865	0.865	0.865	0.824	0.970	0.940		0.949	0.008	0.974	0.949	0.961	0.949	0.998	0.995	Weighted Avg.	0.925	0.025	0.926	0.925	0.925	0.901	0.985	0.962
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Manual Logistic	Time taken to build model: 4.53 seconds === Evaluation on test set === Time taken to test model on supplied test set: 0 seconds === Summary === <table><tr><td>Correctly Classified Instances</td><td>143</td><td>89.375 %</td></tr><tr><td>Incorrectly Classified Instances</td><td>17</td><td>10.625 %</td></tr><tr><td>Kappa statistic</td><td>0.8582</td><td></td></tr><tr><td>Mean absolute error</td><td>0.0684</td><td></td></tr><tr><td>Root mean squared error</td><td>0.207</td><td></td></tr><tr><td>Relative absolute error</td><td>18.2448 %</td><td></td></tr><tr><td>Root relative squared error</td><td>47.8349 %</td><td></td></tr></table>	Correctly Classified Instances	143	89.375 %	Incorrectly Classified Instances	17	10.625 %	Kappa statistic	0.8582		Mean absolute error	0.0684		Root mean squared error	0.207		Relative absolute error	18.2448 %		Root relative squared error	47.8349 %																																																																			
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	Kappa statistic	0.8747							
	Mean absolute error	0.0535							
	Root mean squared error	0.2096							
	Relative absolute error	14.2855 %							
	Root relative squared error	48.4314 %							
	Total Number of Instances	160							
=== Detailed Accuracy By Class ===									
Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	
	0.902	0.008	0.974	0.902	0.937	0.917	0.966	0.929	'(-inf-4.65]'
	0.930	0.068	0.833	0.930	0.879	0.834	0.936	0.826	'(4.65-5.65]'
	0.838	0.024	0.912	0.838	0.873	0.838	0.941	0.878	'(5.65-6.95]'
	0.949	0.025	0.925	0.949	0.937	0.916	0.977	0.922	'(6.95-inf)'
Weighted Avg.	0.906	0.032	0.910	0.906	0.907	0.876	0.955	0.888	
=== Confusion Matrix ===									
a b c d <-- classified as									
37 3 1 0 a = '(-inf-4.65]'									
1 40 1 1 b = '(4.65-5.65]'									
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0 1 1 37 d = '(6.95-inf)'									